MIL-M-24041C
17 August 1981
SUPERSEDING
MIL-M-0024041B(SHIPS)
15 August 1975 and
MIL-M-24041A
20 October 1972
(See 6.6)

MILITARY SPECIFICATION

MOLDING AND POTTING COMPOUND, CHEMICALLY CURED, POLYURETHANE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of two-component, chemically cured, polyurethane compounds for the molding, encapsulating, and potting of watertight electrical connectors, cables, cable end seals, circuit boards, and other electrical components.

1.2 Classification. Compounds covered by this specification shall be of the following categories and types, as specified (see 6.2.1):

Category A - (4, 4' Methylene BIS (2-chloroaniline) cured).
Category B - (Non 4, 4' Methylene BIS (2-chloroaniline) cured).
Type I - Two-component units.
Type II - Premixed, degassed and frozen.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper-Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
UU-T-91 - Tape, Pressure-sensitive Adhesive, Double-coated.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9320, 5970

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MIL-C-915 - Cable and Cord, Electrical, For Shipboard Use, General Specification for.
MIL-R-3065 - Rubber, Fabricated Products.
MIL-W-5086 - Wire, Electric, Polyvinyl Chloride-insulated, Copper or Copper Alloy.
MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-I-7444 - Insulation Sleeving, Electrical, Flexible.
MIL-S-8660 - Silicone Compound.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
A 240 - Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Fusion-Welded Unfired Pressure Vessels.
D 149 - Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies, Test for.
D 150 - A-C Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulating Materials, Tests for.
D 257 - D-C Resistance or Conductance of Insulating Materials, Test for
D 395 - Rubber Property-Compression Set, Test for.
D 412 - Rubber Properties in Tension, Test for.
D 495 - High-Voltage, Low-Current, Dry Arc Resistance of Solid Electrical Insulation, Test for.
D 518 - Rubber Deterioration-Surface Cracking, Test for.
D 624 - Rubber Property-Tear Resistance, Test for.
D 792 - Specific Gravity and Density of Plastics by Displacement, Test for.
D 1149 - Rubber Deterioration-Surface Ozone Cracking in a Chamber (Flat Specimens), Test for.
D 2240 - Rubber Property-Durometer Hardness, Test for.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The compound furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Composition. Before mixing, the compound shall be a two-component, polyurethane system consisting of a prepolymer and a curing agent. The prepolymer shall cure by the addition of the curing agent, and shall not depend on solvent or moisture evaporation for curing. The prepolymer and curing agent shall be as specified herein.

3.2.1 Primer. The use of a primer to improve the adhesion of the compound to the substrate shall be permitted. The primer shall be available from the same manufacturer as the compound with which it is to be used and shall be applied in strict conformance with the manufacturer's recommendations.

3.2.2 Toxicity. The compound shall contain no highly toxic materials, either initially, during application, during or after curing, and shall have no adverse effect on the health of personnel, when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.7.1.1 and 6.2.2).

3.3 Appearance. The liquid mixed compound shall have a uniform homogeneous texture and shall be free of lumps, coarse particles, and air bubbles, preparatory to molding or potting. There shall be no separation of pigment which cannot be readily dispersed. Solidification of only the curing agent shall be permissible, provided it can be reliquefied readily by the application of heat.

3.4 Color. The color of the liquid shall be a consistent uniform texture. Color shall be as specified (see 6.2.1).

3.5 Application properties. The compound shall possess the application properties specified in 3.5.1 through 3.5.4.

3.5.1 Nonvolatile content. Nonvolatile content of the molding compound shall be not less than 99 percent, when tested as specified in 4.7.2.1.

3.5.2 Initial dynamic viscosity. A mixed type I compound shall have a viscosity not greater than 400 poises, nor less than 100 poises, when determined as specified in 4.7.2.2. Type II compound shall have a viscosity not greater than 700 poises, nor less than 250 poises, when determined as specified in 4.7.2.2.

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3.5.3 Application life. Application life of the type I compound shall be not less than 1 hour, when tested as specified in 4.7.2.3.1. Type II compound shall have an application life of not less than 1/2 hour, when tested as specified in 4.7.2.3.2.

3.5.4 Total or ultimate cure time. The Shore "A" hardness of the material shall be not less than 70 when cured for the periods as shown in table I, and when tested as specified in 4.7.2.4.

TABLE I. Total curing time.

Time	Relative humidity	Temperature °F	Requirements
16 hours	50 ± 5 percent	180 ± 2°F	Shore " A" hardness, 70 minimum
3 weeks	50 ± 5 percent	77°F	Shore "A" hardness, 70 minimum

3.5.5 Storage life. Type I material and primer shall meet the requirements of this specification for 12 months from date of manufacture when stored below 80°F in the original unopened containers. Type II materials shall meet the requirements of this specification for 28 days at minus 40°F after receipt of material.

3.6 Mechanical and physical properties. Mechanical and physical properties of the totally cured compound shall be in accordance with table II, when tested as specified in 4.7.3.

TABLE II. Mechanical and physical properties.

Property	Units	Requirements	Test paragraph
Shrinkage	Percent	5 maximum	4.7.3.1
Specific gravity	--	1.0 minimum	4.7.3.2
	--	1.2 maximum	
Shore "A" hardness	--	70-92	4.7.3.3
Tensile strength	Lb/in^2	2,500 minimum	4.7.3.4
Ultimate elongation	percent	400 minimum	4.7.3.5
Tear strength	Lbs/in	150 minimum	4.7.3.6
Compression set	Percent	55 maximum	4.7.3.7
Adhesion:	Lbs/in of width		
To metal		50 minimum	4.7.3.8.1
To rubber		15 minimum	4.7.3.8.2
To polyvinylchloride		15 minimum	4.7.3.8.3
Moisture absorption	Percent	3 maximum	4.7.3.9
Low temperature flexibility	°F	-65 maximum	4.7.3.10

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TABLE II. Mechanical and physical properties. Continued

Property	Units	Requirements	Test paragraph
Heat aging Shore "A" hardness loss at 275°F for 24 hours	--	-7 maximum	4.7.3.11
Property degradation	Percent	50 maximum	4.7.3.12
Fungus resistance	--	Non-nutrient	4.7.3.13
Hydrolytic stability Shore "A" hardness loss	Percent	-20 maximum	4.7.3.14
Ozone resistance	--	No cracking	4.7.3.15

3.7 Electrical properties. Electrical properties of the totally cured molding and potting compound shall be in accordance with table III, when tested as specified in 4.7.4.

TABLE III. Electrical properties. 1/

Property	Units	Requirements	Test paragraph
Resistance to arc	Seconds	50 minimum	4.7.4.1
Dielectric strength	Volts/mil	250 minimum	4.7.4.2
Dielectric constant:			
1 kilohertz at 77°F	--	10 maximum	4.7.4.3
1 megahertz at 77°F	--	8 maximum	4.7.4.3
Power factor:			
1 kilohertz at 77°F	--	0.08 maximum	4.7.4.3
1 megahertz at 77°F	--	0.10 maximum	4.7.4.3
Volume resistivity	Ohm - CMS		
at 77°F		4×10^{11} minimum	4.7.4.4
at 250°F		2.5×10^{9} minimum	4.7.4.4
Surface resistivity	Ohms		
at 77°F		2.4×10^{12} minimum	4.7.4.4
at 250°F		2.4×10^{9} minimum	4.7.4.4
Insulation resistance:	Megohms		
Initial at 77°F		300,000 minimum	4.7.4.5
at 250°F		100 minimum	4.7.4.5
After hydrolytic stability test		1000 minimum	4.7.3.14.1
High potential		No breakdown	4.7.4.6
Flame resistance overload		Shall not ignite	4.7.4.7

1/Based on average compound properties without special processing.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command. Qualification inspection shall consist of the examination and tests specified in 4.7.

4.3.1 Samples for qualification tests. Qualification test samples shall consist of at least 5 quarts of base compound, together with the necessary curing agent and primer. The base compound and curing agent shall be furnished in containers of the type to be used in filling contract orders. After successfully conforming to all the requirements of this specification, approval of the material will be granted, and will be identified by reference to the manufacturer's code or formula number. Samples shall be identified as follows:

SAMPLES FOR QUALIFICATION TESTS

Molding and Potting Compound, Chemically Cured, Polyurethane.

Specification MIL-M-24041.
Category No.
Type No.

Name of manufacturer and address of plant which produced qualification sample.

Manufacturer's code no. or formula no.

Manufacturer's code no. or formula no. of primers for adhesion tests.

Submitted by (name) (date) for qualification tests in accordance with the requirements of Military Specification MIL-M-24041 under authorization (reference authorizing letter).

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Date of manufacture.

Mixing instructions.

Manufacturer's data.

4.3.2 Manufacturer's data. Two copies of the manufacturer's instructions for the mixing and application of the compound, to be identical in content to those instructions required in 5.1.3 and two copies of the manufacturer's test report, containing data showing that material submitted for qualification conforms to the qualification requirements of this specification, shall be submitted simultaneously with qualification samples.

4.4 Quality conformance tests.

4.4.1 Inspection lot. A lot shall consist of all compound manufactured as one batch and offered for delivery at one time.

4.4.2 Sampling.

4.4.2.1 Sampling for inspection of material. Enough containers shall be selected at random to form two separate specimens. These containers shall be sealed, marked, and held for testing of the material. Specimens shall be subjected separately to the following tests. If either specimen fails one or more of these tests, this shall be cause for rejection of the lot.

(a) Examination of product (see 4.7.1)
(b) Viscosity (see 4.7.2.2)
(c) Application life (see 4.7.2.3)
(d) Cure time (at 180°F only) (see 4.7.2.4)
(e) Hardness (see 4.7.3.3)

4.4.2.2 In addition, the product shall be subjected to any of the other tests specified herein which the contracting activity considers necessary to determine conformance with the requirements of this specification.

4.5 Government production control tests. Government production control tests as considered desirable may, at the option of the contracting activity, be conducted on samples selected from a batch of molding compound. Samples shall be furnished by the manufacturer.

4.6 Test conditions.

4.6.1 Test specimen preparation.

4.6.1.1 Mixing of type I molding material. Type I material curing agent may solidify when subjected to temperatures below 65°F. When this condition exists, liquefy the curing agent in accordance with manufacturer's instructions. After liquefication is complete, allow the curing agent to cool to 77°F ± 2°F. The two curing agents shall be thoroughly mixed in the proper proportions in either a mixing machine designed for this application, or in a metal, polyethylene, or glass container having approximately twice the volume of material. Mixed material shall be degassed under a vacuum in a period of less than 20 minutes.

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4.6.1.2 Thawing of type II material. Premixed, frozen cartridges of molding compound shall be thawed in accordance with the manufacturer's instructions, when taken from a storage temperature of at least minus 40°F, prior to testing.

4.6.1.3 Molding and potting. Molds for disc specimens and special shapes as indicated in individual test descriptions shall be of polyethylene or of a Teflon-coated metal or equal sprayed with a silicone release agent or of any other composition having a nonadhesive surface which will not transfer to the test specimen. The mold shall be constructed to facilitate easy removal of the specimen. Panels of the size and composition indicated in individual test descriptions shall be designed for coating by suitable mold insertion or coating technique. Panels shall be thoroughly cleaned with a suitable solvent and dried, prior to the application of the primer and molding compound. Molding compound shall be poured or extruded slowly into the mold until a slight excess is obtained. The mold shall then be covered with a flat, smooth plate and a pressure not to exceed 10 pounds per square inch (lb/in^2) applied.

4.6.1.4 Curing. Each specimen prepared in accordance with 4.6.1.1 and 4.6.1.2 shall be cured for 16 hours at 180°F ± 5°F, unless otherwise specified herein. Cured specimens shall be kept at standard conditions until tests commence. Tests on cured specimens shall commence not later than 24 hours after completion of the specified cure.

4.6.1.5 Finished specimens. Surfaces of finished specimens shall be clean, smooth, free from holes and bubbles, and all corners and edges shall be sharp.

4.7 Test methods.

4.7.1 Examination of product. The compound shall be examined carefully to determine conformance with this specification with respect to requirements for which no test methods are specified herein.

4.7.1.1 Certification data/report. The contractor shall furnish toxicological data and formulations required to evaluate the safety of the material for the proposed use. A certification/data report to this effect shall be provided in accordance with the data ordering document included in the contract (see 6.2.2).

4.7.2 Application property tests.

4.7.2.1 Nonvolatile content. Five to 10 grams (g) of the mixed molding compound shall be transferred as soon after mixing as possible to a tared dish about 8 centimeters (cm) in diameter. A tight fitting tared cover shall immediately be placed over the dish and weight determined to the nearest milligram (mg). The cover shall then be removed and the sealing compound heated for 24 hours at 200°F ± 10°F. It shall then be cooled in a desiccator, the cover replaced, and weight determined to the nearest mg. The percent nonvolatile content shall be calculated as follows:

$$\text{Percent nonvolatile content} = \frac{\text{final weight of compound}}{\text{initial weight of compound}} \times 100$$

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4.7.2.2 Viscosity. Viscosity determination on the mixed type I material or freshly thawed type II material shall be made with the Brookfield viscosimeter model RVF (or equal) with a no. 6 spindle operated at 10 revolutions per minute (r/min). Readings shall be taken when the pointer first assumes a steady position after release of the clutch. The material and the viscosimeter shall be kept at a uniform temperature of 77°F ± 2°F during the test.

4.7.2.3 Application life.

4.7.2.3.1 Type I materials. A 250-g sample of mixed compound shall be used to determine application life, which shall commence at the end of the mixing period. A standard 1/2 pint can with its retaining flange removed shall be used in conjunction with a Brookfield viscosimeter model RVF (or equal) equipped with a no. 7 spindle and operated at 10 r/min. Viscosity shall be determined at the end of a 1-hour period. Supplemental readings shall be made at 10-minute intervals until a reading of 2,500 poises is attained which shall be considered as the end of the application life. The spindle shall not be drawn from the material during the test. Readings shall be taken when the pointer on the viscosimeter dial first assumes a steady position after a minimum of three revolutions.

4.7.2.3.2 Type II materials. A cartridge of molding compound, thawed in accordance with 4.6.1.2, shall be attached with a Pyles or Semco No. 440 (or equal) nozzle. The Pyles or Semco gun shall be attached to a constant supply of compressed air at 75 lb/in^2. Three or 4 inches of material shall be extruded initailly to fill the nozzle and to remove any trapped air. After a period of exactly 1/2 hour, measured from the end of the thawing period, the material shall be extruded into a tared container for 1 minute and the weight of the extruded material determined. The end of the application life shall be when the extrusion rate is less than 30 g per minute. The cartridge of material shall be kept at 77°F ± 2°F throughout the test.

4.7.2.4 Cure time. Compound specimens having 3 by 3 inches minimum face dimensions and a minimum thickness of 1/4 inch shall be prepared as specified in 4.6.1. Hardness shall be determined at the expiration of the curing period with a type Shore A durometer in accordance with ASTM D 2240, using a 3-second application time.

4.7.3 Mechanical and physical property tests.

4.7.3.1 Shrinkage. A cubical mold, approximately 1 inch on each side and having an open top shall be constructed in accordance with 4.6.1.3. Its volume at 77°F ± 2°F and relative humidity (R.H.) 50 ± 5 percent shall be determined. A compound specimen shall be prepared in the mold and cured in accordance with 4.6.1.4. After curing, the specimen

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shall be placed in a circulating air oven at 180°F ± 5°F for 96 hours. It shall then be removed, cooled, and its volume at 77°F ± 2°F determined by the water displacement method. Percent shrinkage shall be calculated as follows:

$$\text{Percent shrinkage} = \frac{V_1 - V_2}{V_1} \times 100$$

Where:

V_1 = volume of mold
V_2 = final volume of compound

4.7.3.2 Specific gravity. The cured test specimen shall be tested in accordance with ASTM D 792.

4.7.3.3 Hardness. Compound specimens having 3 by 3 inches minimum face dimensions and a minimum thickness of 1/4 inch shall be prepared as specified in 4.6.1. Hardness shall be determined with a type A Shore hardness durometer in accordance with ASTM D 2240, using a 3-second application time.

4.7.3.4 Tensile strength. Tensile strength specimens shall be cut from a sheet of the compound 0.080 ± 0.010 inch thick, molded in a smooth surface mold in accordance with 4.6.1. Test specimens shall be cut using die C and tested in accordance with ASTM D 412.

4.7.3.5 Ultimate elongation. The test specimen shall be the same as described in 4.7.3.4 and shall be tested in accordance with ASTM D 412.

4.7.3.6 Tear strength. Tear strength specimens shall be cut from a sheet of the compound 0.080 ± 0.010 inch thick, molded in a smooth surface mold in accordance with 4.6.1. Test specimens shall be cut using die C and tested in accordance with ASTM D 624.

4.7.3.7 Compression set. A cured compound specimen 0.500 ± 0.005 inch thick and 1.129 ± 0.001 inch in diameter shall be molded in accordance with 4.6.1. Compression set shall be determined in accordance with ASTM D 395, method B. The test shall be conducted at 158°F for 22 hours.

4.7.3.8 Adhesion.

4.7.3.8.1 Adhesion of compound to metal panel.

4.7.3.8.1.1 Preparation of specimens. The surface of the metal panels listed in table IV shall be vapor degreased, sand blasted, and again vapor degreased. The cleaned surface shall be primed with a thin coat of the recommended primer in accordance with the manufacturer's instructions and allowed to dry. A quantity of molding compound shall be freshly mixed in accordance with 4.6.1 to form a coating having a thickness of 1/8 inch ± 0.030 inch, and applied to the primed panel. A 3-inch by 6-inch area of a 3-inch by 12-inch strip of 60-mesh stainless screen shall be intimately coated with the freshly mixed molding compound and placed on the coated area of the panel, leaving a loose end 6 inches long. The panel shall be cured in accordance with 4.6.1.4.

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TABLE IV. Panels for adhesion test.

Composition	Specification	Size	Number of specimens cut from each panel assembly
		Inches	
CRES	Type 316 of ASTM A 167 or ASTM A 240	3 by 6 by 1/16	2
Monel	QQ-N-281, class A		
Rubber (neoprene)	MIL-R-3065	3 by 6 by 1/8	2
Polyvinyl chloride	MIL-I-7444	3 by 6 by 1/8	2

4.7.3.8.1.2 Immersion of the metal panel. Immediately after completion of curing, the metal panel shall be immersed vertically for 7 days in a closed glass vessel containing a 3 percent aqueous sodium chloride solution, with 4 inches of the panel exposed to the test fluid and the remainder of the panel exposed to the air vapor mixture. Temperature during the test shall be maintained at 158°F ± 2°F. At the expiration of the test period, the fluid shall be cooled to 77°F ± 2°F. The panel shall then be removed and adhesion determined in accordance with 4.7.3.8.1.3 within 10 minutes after removal from the fluid.

4.7.3.8.1.3 Testing adhesion of compound to metal panel. Panels prepared and conditioned in accordance with 4.7.3.8.1.1 and 4.7.3.8.1.2 shall be tested as follows: Two 1-inch wide strips shall be cut lengthwise through the screen and compound to the panel surface and extended the full length of the loose end of the screen. Edges of the panel shall not be used as one edge of the test strip. The panel shall be tested individually in an autographic testing machine with a capacity such that the tension at failure is not more than 85 percent nor less than 15 percent of the full scale load. Rate of separation of the jaws shall be 2 inches per minute and shall be uniform at all times. Specimens shall be mounted in the machine so that the loose end of the 1-inch wide screen strip will be folded 180 degrees as it is pulled from the panel. Each strip shall be pulled as follows: A cut through the compound to the panel at the junction of separation shall be made at an angle of 45 degrees in the direction of separation. If a cohesive failure develops or if the compound separates from the screen, similar 45-degree cuts shall be made as necessary to promote separation of the compound from the panel. Adhesion in pounds shall be automatically recorded on a chart as a continuous curve. Adhesion value shall be calculated by averaging the maximum and minimum forces required to separate the compound from the panel. If cohesive failures occur, the adhesion value shall be reported as greater than the measured value.

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4.7.3.8.2 Adhesion of compound to neoprene.

4.7.3.8.2.1 Preparation of specimens. Two 3-inch by 6-inch by 0.125-inch thick sheets of rubber conforming to MIL-R-3065, and used for cables conforming to MIL-C-915 shall be abraded on one side with 80 grit sandpaper and wiped clean with dry gauze. Rubber sheets with abraded side up shall be supported by 3-inch by 6-inch by 0.032-inch thick metal sheets attached to the unabraded side with double-faced pressure sensitive tape conforming to UU-T-91. A thin coat of the recommended primer shall be applied to the abraded surfaces in accordance with the manufacturer's instructions, and allowed to dry. Primed surfaces of rubber sheets shall be placed facing each other, separated by two 1/8-inch thick spacers along the 6-inch edges. The two sheets shall be taped together along the 6-inch edges and one of the 3-inch edges. Extrude freshly mixed molding compound into the cavity filling it to within 1 inch of the top of the assembly. Cure material in accordance with 4.6.1.4 and cut two 1-inch by 6-inch strips from the test specimen assembly. Outside cuts shall be parallel to and approximately 1/2 inch from the lengthwise edge of the assembly.

4.7.3.8.2.2 Testing of adhesion of compound to neoprene. Loose ends of the rubber specimen which are left by filling the assembly to within 1 inch of the top in 4.7.3.8.2.1 shall be attached to opposite grips of an autographic tensile testing machine with a capacity such that tension at failure shall not be more than 85 percent nor less than 15 percent of the full scale. Rate of grip separation shall be 2 inches per minute and shall be uniform at all times. Adhesion in pounds shall be automatically recorded on a chart as a continuous curve for a minimum of 6 inches of grip separation. Adhesion value shall be calculated by averaging the maximum and minimum forces required to separate the compound from the rubber.

4.7.3.8.3 Adhesion of compound to polyvinyl chloride.

4.7.3.8.3.1 Preparation of specimens. Two 3-inch by 6-inch by 0.125-inch thick sheets of polyvinyl chloride conforming to MIL-I-7444 shall be prepared for priming by applying methyl ethyl ketone to one side of each strip to make the surface tacky. A thin coat of recommended primer shall be applied to the tackified surface in accordance with the manufacturer's instructions and allowed to dry. Primed surfaces of the polyvinyl chloride sheets shall be placed facing each other separated by two 1/8-inch thick spacers along the 6-inch edges and one of the 3-inch edges. Extrude freshly mixed molding compound into the cavity filling it to within 1 inch of the top of the assembly. Cure material in accordance with 4.6.1.4 and cut two 1-inch by 6-inch strips from the test specimen assembly. Outside cuts shall be parallel to and approximately 1/2 inch from the lengthwise edge of the assembly.

4.7.3.8.3.2 Testing of adhesion of compound to polyvinyl chloride. Polyvinyl chloride specimens shall be tested in accordance with 4.7.3.8.2.2.

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4.7.3.9 Moisture absorption. A specimen, 1.0 inch by 1.0 inch by 1.0 inch, shall be molded in accordance with 4.6.1. The specimen shall be weighed and immersed for 24 hours in a distilled water bath held at 200°F ± 5°F. The specimen shall be removed, blotted dry and weighed. The percent of water absorption shall be calculated as follows:

$$\text{Percent } H_2O \text{ absorption} = \frac{\text{Final weight - initial weight}}{\text{initial weight}} \times 100$$

4.7.3.10 Low temperature flexibility. Three test panels of stainless steel conforming to type 316 of ASTM A 167 or ASTM A 240, measuring 1 inch by 6 inches by 0.032 inch shall be coated on one side with the recommended primer in accordance with the manufacturer's instructions. A quantity of molding compound shall be mixed in accordance with 4.6.1 and applied to the primed surfaces in a manner which will produce cured coatings with a thickness of 0.050 inch to 0.060 inch, leaving 1 inch at each end of the panels uncoated. Immediately after completion of curing, the test panel shall be immersed in a 3 percent aqueous sodium chloride solution for a period of 3 weeks (504 hours ± 12 hours). The test specimens shall then be inserted in a flexibility jig as shown on figures 1 and 2, so that the uncoated side will contact the contour block and the weight will contact only the uncoated end of the panel. The flexibility jig and test specimens shall be subjected to a conditioning temperature of - 65°F ± 2°F for 6 hours. After the specified conditioning, the specimens shall then be bent around the curved portion of the flexibility test jig by releasing the fastening hook. Test specimens shall be removed and examined for failure due to blistering, cracking, or loss of adhesion.

4.7.3.11 Heat aging. Compound specimens having a 3-inch by 3-inch minimum face dimension and a minimum thickness of 1/4 inch shall be prepared in accordance with 4.6.1 and cured for 16 hours at 180°F ± 2°F. Specimens shall be measured for hardness using a type A Shore hardness durometer in accordance with 4.7.3.3, and then placed in a circulating oven at 275°F ± 2°F and measured for loss of Shore "A" hardness.

4.7.3.12 Property degradation. Compound specimens prepared in accordance with 4.7.3.4 shall be placed in water at 158°F ± 2°F for a period of 2 weeks (336 hours ± 12 hours). Remove and allow to dry at room temperature (77°F ± 2°F) and relative humidity of 50 percent for a period of one week. Tensile strength reduction shall be less than 50 percent when compared to the specimen previously tested in accordance with 4.7.3.4.

4.7.3.13 Fungus resistance. Cured specimens prepared in accordance with 4.6.1 shall be tested as specified in MIL-E-5272.

4.7.3.14 Hydrolytic stability test. Aluminum rings, 1-3/8 inch inside diameters, shall be cut into 1-inch lengths. Rings shall be cleaned by scrubbing and rinsing with soap-free and grease-free rags or paper towels, using methyl ethyl ketone. Rings shall be wiped dry rather than allowing the solvent (MEK) to evaporate. A primer specified by the manufacturer shall be applied to the inside of four of the aluminum rings, and dried in accordance with the manufacturer's instructions. Two of the rings

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shall be cast with two brass electrodes set at equal distances from the aluminum ring at 3/4-inch centers. Electrodes shall be 1/16-inch diameter, and approximately two inches in length. Specimens shall be cast so that 1/2 inch of each electrode is imbedded into the polyurethane. These shall be used to obtain insulation resistance values. Two other rings shall be cast with polyurethane, and shall be used to obtain shore "A" hardness values. Mixing, degassing, casting and curing of the specimens shall be in accordance with 4.6.1.

4.7.3.14.1 Four test specimens shall be placed in a humidity chamber which maintains 95 percent relative humidity at 160°F ± 2°F for 120 days. After 120 days, the specimens shall be removed, and cooled for 4 hours at 77°F ± 2°F and 50 ± 5 percent relative humidity. Insulation resistance shall be determined on the specimens with the brass electrodes using a megohm bridge, at a potential of 500 volts (V). Electrification time shall not exceed 2 minutes. Hardness readings shall be determined on the remaining two specimens, using a type A Shore hardness durometer. Each determination shall be the average of 5 readings.

4.7.3.15 Ozone resistance. An ozone resistance test, of 28-day duration at 50 parts per hundred million ozone, 104°F in accordance with ASTM D 518, procedure B and ASTM D 1149 shall be conducted. There shall be no evidence of ozone cracking in the test specimen.

4.7.4 Electrical property tests.

4.7.4.1 Resistance to arc. Three disc specimens, 4 inches in diameter and 1/8 inch thick, shall be prepared as specified in 4.6.1. Arc resistance test shall be conducted in accordance with ASTM D 495.

4.7.4.2 Dielectric strength. Three disc specimens, 4 inches in diameter and 1/8 inch thick shall be prepared as specified in 4.6.1. Test shall be conducted in accordance with ASTM D 149. The tests shall be made under oil at a frequency not exceeding 100 hertz (Hz). The voltage shall be increased uniformly at the rate of 500 volts per second (V/s).

4.7.4.3 Dielectric constant and power factor. Three disc specimens, 4 inches in diameter and 1/8 inch thick, shall be prepared as specified in 4.6.1. Tests shall be conducted in accordance with ASTM D 150. Lead or tin foil electrodes shall be used and applied to the specimen with silicone grease conforming to MIL-S-8660. The electrodes shall consist of two lead or tin foil discs, one of the same diameter as the specimen and the other 2 inches in diameter, centrally located on the opposite face of the specimen. Test current shall be introduced to the foil through two brass discs. The upper disc shall be 2 inches in diameter by 1 inch in thickness, and the lower disc shall be 4 inches in diameter and 1 inch thick. The specimens shall be tested at frequencies of 1 kilohertz and 1 megahertz at a temperature of 77°F ± 2°F and a relative humidity of 50 ± 5 percent. Calculation shall include corrections for edge and ground capacitance effects.

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4.7.4.4 Volume and surface resistivity. Three disc specimens, 4 inches in diameter and 1/8 inch thick, shall be prepared as specified in 4.6.1. Tests shall be conducted in accordance with ASTM D 257 using a General Radio Type 544 B bridge, or equal instrument, with a test voltage of 500 V. Readings shall be made 1 minute after application of current. Lead or tin foil electrodes shall be applied to the specimen with silicone grease conforming to MIL-S-8660. The guarded electrode shall be a disc, 2 inches in diameter, centrally located on one face of the specimen. The guard electrode shall be a concentric ring of 2-9/32 inches inside diameter with an outside diameter equal to that of the specimen. The unguarded electrode shall be a foil disc 4 inches in diameter applied to the opposite side of the specimen. Test current shall be introduced to the guarded electrode, the guard electrode, and the unguarded electrode by means of a brass disc 2 inches in diameter by 1 inch in thickness, a brass ring 2-5/16 inches inside diameter by 4 inches outside diameter by 1/8 inch thick, and a brass disc 4 inches in diameter, respectively. Calculations necessary for volume and surface resistivity shall be made using the approximate ASTM formulae for effective area and effective perimeter. All specimens shall conform to minimum requirements. Tests shall be conducted at 77°F ± 2°F and 250°F ± 2°F after a conditioning period of 30 minutes at test temperature.

4.7.4.5 Insulation resistance. Three test specimens shall be molded in accordance with 4.6.1 to the dimensions specified on figure 3. The mold in which the specimens are cast shall provide for the accurate spacing of brass rod electrodes. The electrodes shall be 1/16 inch diameter and approximately 3 inches long, and shall be inserted at the opposite end of the rectangular specimen in such a manner that 1-1/2 inches of the length of each pair shall be spaced 0.050 ± 0.010 inch apart. Four pairs of electrodes shall be spaced at least 0.750 inch apart. Measurements shall be made by the megohm bridge, using a potential of 500 V. Electrification time shall not exceed 2 minutes. Tests shall be conducted at 77°F ± 2°F and 250°F ± 2°F after a conditioning period of 30 minutes at test temperature.

4.7.4.6 High potential resistance. Cast assemblies as specified in 4.7.4.5 shall be utilized in this test. A potential of 2,000 volts root mean square (Vrms), 60 hertz, shall be applied to all contacts for a period of 1 minute. The test voltage shall be applied gradually at the rate of approximately 500 V each second until the specified voltage of 2,000 V is reached. This test shall be conducted on assemblies as initially prepared. No breakdown shall be allowed.

4.7.4.7 Flame resistance. A test specimen shall be prepared as specified in 4.6.1. The specimen shall consist of a cylinder of compound 1-1/2 inches long and 1/2 inch in diameter. A 12-inch length of size no. 16 insulated wire conforming to MIL-W-5086 shall be molded into the center of the specimen along its lengthwise axis. The specimen shall be suspended taut in a horizontal position. An electric current of 55 amperes direct current (d.c.) shall be passed through the wire for 2-1/2 minutes, and the material examined for ignition and charring.

4.7.5 Packaging inspection. Packaging, packing and marking shall be inspected to determine conformance with the requirements of section 5.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Unit packaging (see 6.2.1).

5.1.1 Type I material. The base compound and correct amount of curing agent shall be packaged in individual containers. Each base compound container shall be packaged with one curing agent container with a fiberboard separator between the two containers in a manner which will prevent accidental separation but which will permit easy separation for mixing purposes. Two compartment unit packs are acceptable providing positive separation of compound and curing agent is achieved.

5.1.2 Type II material. Type II compound shall be supplied premixed, degassed and quick frozen in 2-1/2-, 6-, and 12-ounce plastic cartridges, complete with plunger and cartridge cap or plug, as manufactured by Pyles Industries, Inc. and Semco Research Co., or equal.

5.1.3 Printed sheet instructions for proper use of compound, preparation of surfaces (metal, plastics, rubber, neoprene, etc.) and proper use of primer shall be furnished with each unit package of compound.

5.2 Packing. All exterior shipping containers in a shipment shall contain the same number of the type of unit packages. Shipping containers shall be of uniform size and shall be designed to insure that damage is prevented during handling and shipping. Frozen shipments shall be packed in a manner that will prevent thawing or other damage during transit.

5.3 Marking. In addition to any special marking required by the contract (see 6.2.1), unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with MIL-STD-129. Marking shall include specification number, type, and color; name of manufacturer; date of manufacture and expected life expiration date (month and year for type I material and month, day, and year for type II material) and recommended storage temperature, and any precautionary instructions normally required.

6. NOTES

6.1 Intended use of polyurethane. Compounds covered by this specification are flexible cold flow resistant materials with excellent electrical properties for use in sea water. These compounds will be used for the sealing and reinforcement of electrical connectors, wiring, and other electrical apparatus, and for their protection against corrosion or contamination. Compounds covered by this specification have been electrically tested to 1 megahertz. Compounds qualified to category B are not interchangeable with category A materials. It is recommended that where a material switch is contemplated, the material be adequately reviewed by cognizant personnel to preclude undesirable results. Some category B compounds are noted to have an objectionable odor during and after curing.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification
(b) Category and type of material (see 1.2).
(c) Color required (see 3.4).
(d) Required size (see 5.1).
(e) Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in a contract which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID	Option
4.7.1.1	Certification/Data Report	UDI-A-23264	-----

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List QPL 24041 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

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6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement as cited in ABC-NAVY-STD-17. When amendment, revision or cancellation of this specifition is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.6 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

International interest (see 6.5)

Custodians:
Army - MR
Navy - SH
Air Force - 11

Preparing activity:
Navy - SH
(Project 9320-0249)

Review activities:
DLA - GS
Army - AT, MI
Navy - EC

User activities:
Navy - AS, CG

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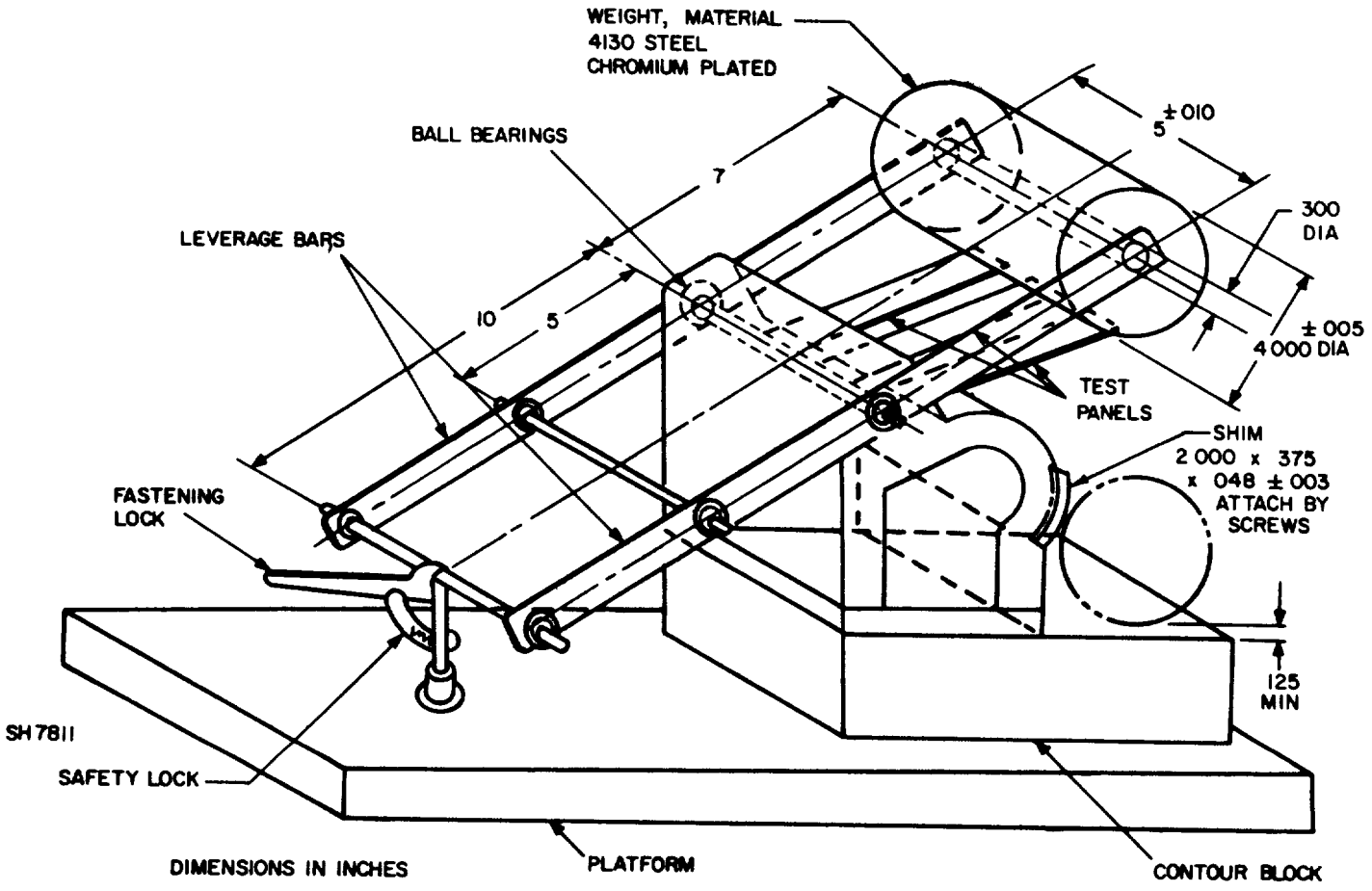


FIGURE 1. Low temperature flexibility apparatus.

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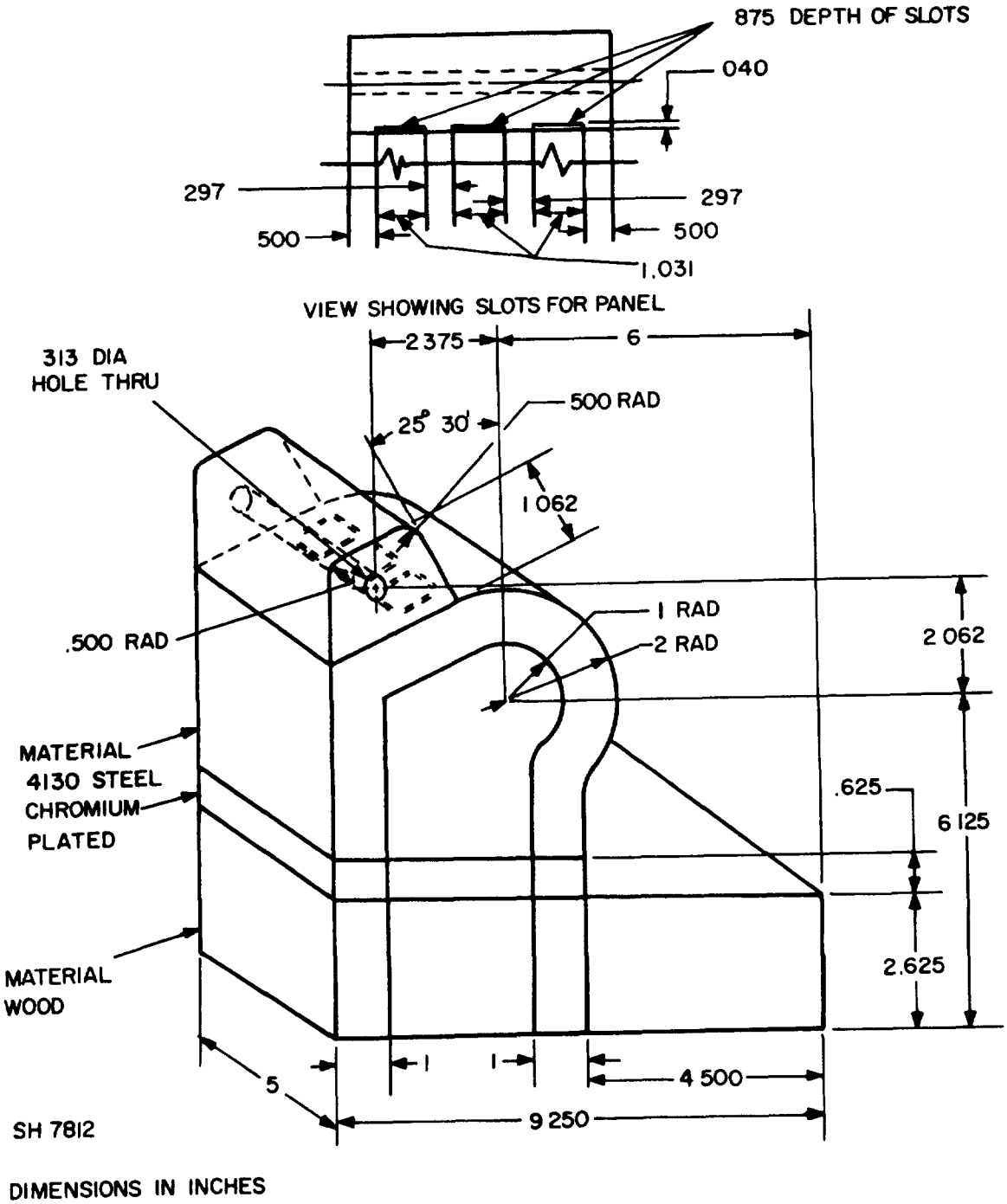
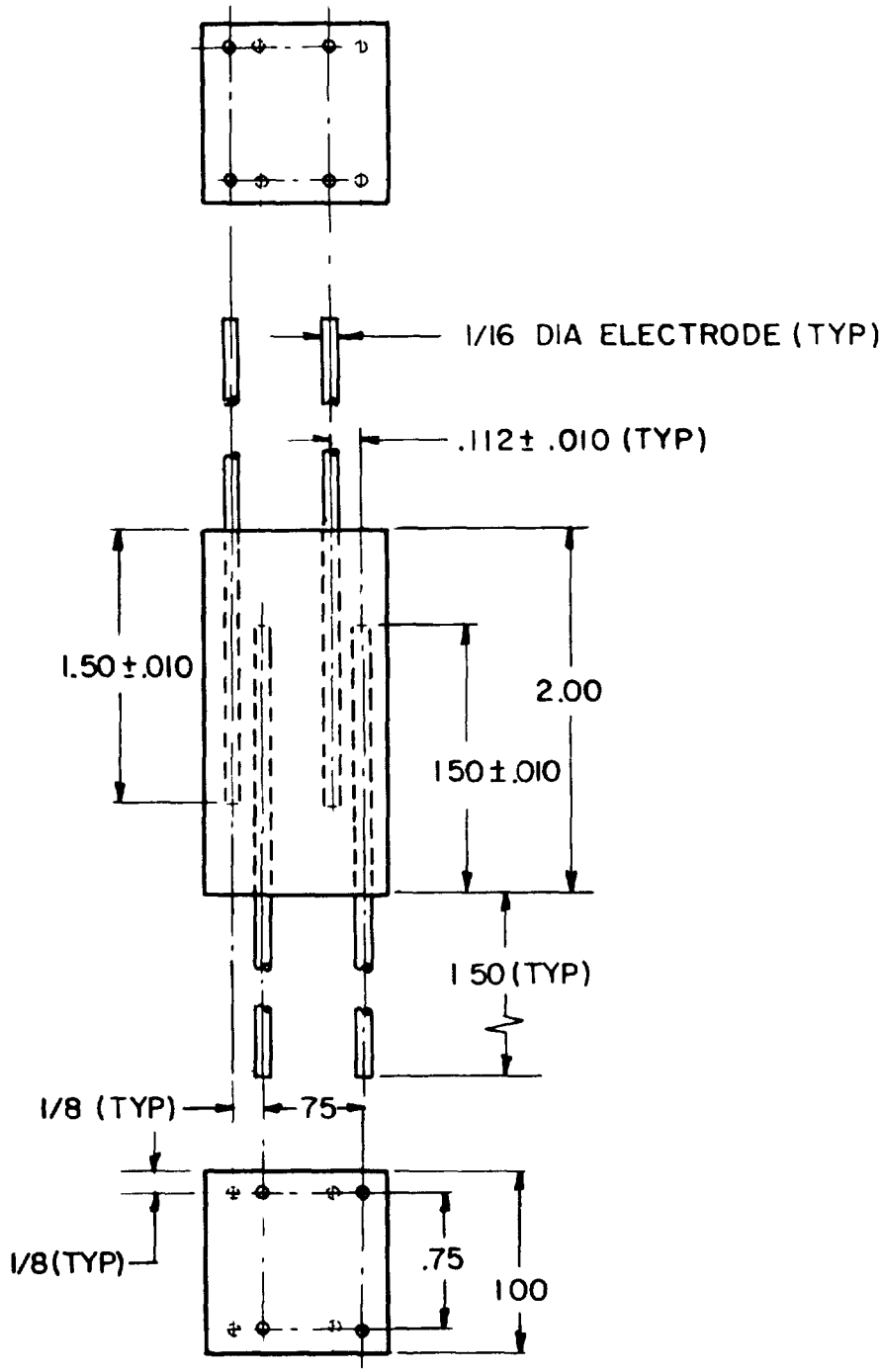


FIGURE 2. Contour block.

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SH 7813

FIGURE 3. Insulation test specimen.

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OMB Approval
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B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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4 REMARKS

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